YATHARTH GOYAL

SOFTWARE ENGINEERING · BACKEND SYSTEMS · JAVA

Jaipur, Rajasthan, India | yatharthgoyal657@gmail.com | GitHub | LinkedIn | LeetCode | Portfolio code

TECHNICAL SKILLS

Languages Java, JavaScript, Python, SQL, HTML, CSS
Frameworks & Libraries React, Vite, Node.js, Express.js, FastAPI, REST APIs
Databases PostgreSQL, PostGIS, SQLite, Redis / BullMQ
Cloud & Infrastructure Docker, Render, Neon, AWS SAM (deployment templates)
Developer Tools Git, GitHub Actions, Playwright, pytest; Java DSA: 16-topic practice repository
Machine Learning & AI scikit-learn, YOLO, pandas, NumPy, SciPy / HiGHS

SELECTED PROJECTS & ENGINEERING CONTRIBUTIONS

AegisED | Emergency Operations & Decision Support · LIVE DEMO | SOURCE | DEMO

React · Express · PostgreSQL · FastAPI · scikit-learn · Render

- Engineered transactional intake, doctor/bed allocation, treatment and discharge flows; used PostgreSQL row locks and rollback to protect resource consistency under concurrent requests.
- Implemented backend RBAC, session safeguards and audit logging; connected an Express gateway to a human-reviewed FastAPI ML service with override tracking.
- Compared four triage model candidates; documented 0.789 macro-F1 on 900 held-out synthetic cases (non-clinical), alongside API and Python validation.

EvalForge | Evidence-Based Assessment Intelligence · PRIVATE · NOT DEPLOYED

React · Express · PostgreSQL · Redis/BullMQ · FastAPI · Docker

- Developed versioned assessment rubrics, criterion-linked evidence, instructor approvals/overrides, appeals and organization-scoped access checks.
- Designed durable asynchronous grading: PostgreSQL job persistence, Redis/BullMQ dispatch, idempotent writes, bounded retries and dead-letter recovery.
- Documented 15/15 backend and 7/7 evaluator tests in local checks; production deployment, hosted CI and full real-service E2E remain pending.

Grid Flex | Industrial Energy Scheduling Optimizer · LOCAL PROTOTYPE | SOURCE

Python · SciPy/HiGHS · MILP · JavaScript · AWS SAM

- Formulated a mixed-integer scheduling model for five synthetic production jobs, solar supply, battery dispatch and grid limits; computed cost, balanced and carbon-priority plans.
- Implemented independent post-solve audits for power balance, state-of-charge, deadlines and reported costs; added outage and solar-stress analysis.
- Passed 52/52 documented local regression tests and desktop/mobile UI checks; AWS Lambda/API Gateway/DynamoDB integration is templated, not deployed.

RouteGuard 2.0 | Road-Hazard Intelligence · IN PROGRESS

Python · YOLO · React/Leaflet · Express · PostGIS · TomTom

- Developing hazard-informed route comparison with a React/Leaflet driver interface, Express APIs, PostGIS and TomTom routing; conducting dataset-label audits for road-damage detection.
- Research target (not achieved): ≥95% precision and recall per severe-hazard class on independent test data; experiments are not production safety claims.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech), Information Technology | JECRC, Jaipur | Expected 2028
CGPA: 9.00/10.00
Certifications Credentials withheld pending issuer-issued verification; no exam passes claimed.